

Spatial Informatics (GS2.401)

Lect 1: Introduction

31/07/2026

Dr. RC Prasad and Dr. Kuldeep Kurte

“Everything is Spatial”

Course Structure

Modules	Topics
Pre-Quiz1	• Introduction • Spatial Data and it's applications • Collection, error corrections
Pre-MidSem	• Spatial data analysis and operations • Maps, Co-ordinate systems and projections
Pre-Quiz2	• Spatial Tessellation, autocorrelations, interpolation • WebGIS, Standardization • Network analysis
Pre-EndSem	• Terrain analysis • Hydrology analysis • LiDAR processing • Cartographic generalization • Applications and Advanced topics

Types of Evaluation	%weightage
Course project • Mid-review • End review	25% – 30%
Homework/Assignments + Quiz (2)	25% – 30%
Midsem Exam	15% – 20%
Endsem Exam	25% – 30%

**Further, Spatial Data Science course is offered Spring Sem
(with SI as a pre-requisite)**

Teaching-Learning Strategies

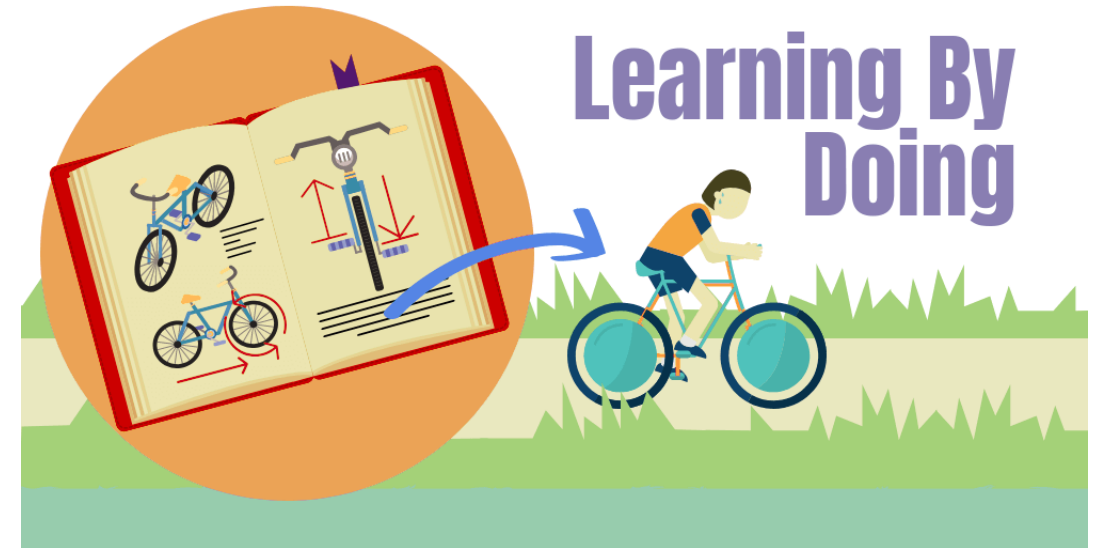
Teaching – Learning

- Lectures
- Guest Lectures
- Reading research papers
- Class participation in Q&A, discussions
- Online discussions over MS Teams

Hands on exercises – in class and assignments

- Short Presentation and Discussion led by Student.
- Course project on conceptualization and implementation
- Real world applications
- Multi-disciplinary approach

Philosophy:



<https://paralearning.in/experiential-learning-learning-by-doing/>

Expectations and Ground Rules

- This course will be largely practical/CW hands-on (more weightages)
- Almost each class will have some form of hands-on
 - Hands-on followed by discussion to understand the concepts better
 - CW will have hard deadlines, mostly on same day or before next class
 - Late submission policy:
 - Late by 1 day: 80% reduction in the total marks
 - Late by 1 week: 50% reduction in the total marks
 - Late by > 1week considered as a no submission
- Exam no show:
 - Re-examination
- All communications must made through Moodle if possible so that it will be visible to all, else via email if required.
- Do not mess-up with the course TAs, they are here to facilitate the smooth functioning of the course

- To make it fair to those who submit the assignments on time
- Please convey the reason upfront

Plagiarism/ Mal-practice

- **Copying other's answers/maps/outputs:**

- Both source and destination will lose total marks if you are caught.
- You can help your friends/colleagues to understand the concepts but do not help them cheat
- It is okay to lose some marks by submitting late but do not cheat with your conscience by doing such mal-practices

- **GenAI:**

- Mostly it won't be useful.
- We are not going to write any essay in the CW

Lab assignment submission

General instructions:

- Lab assignment must be submitted via Moodle link by the end of the class
- Take a screenshot of various step you follow to generate specific output, add some heading and paste it in word
- Finally create pdf and upload it.
- More specific instructions will be provided along with each assignment

*** Details of Assignments/Projects will be announced during the course.**

Location Data in Our Daily Life

Location is an invisible part of everything we do every day



Almost every digital service answers one of three questions:



Where am I?



What is around me?



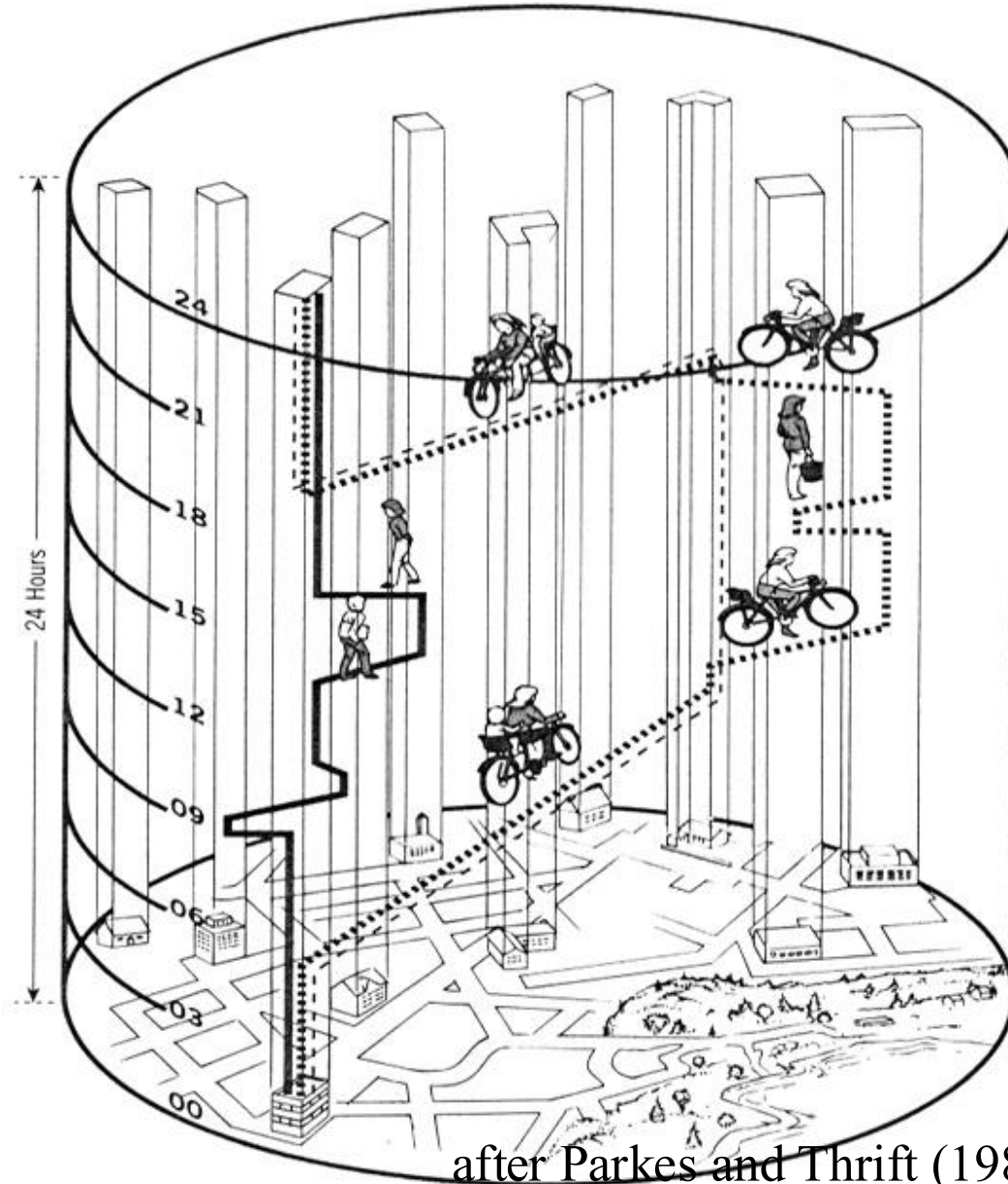
How do I get there?



These simple questions are the foundation of **GIS** and **Spatial Science**.

“... space by itself and time by itself, are doomed to fade away into mere shadows, and only a kind of union between the two will preserve an independent reality” (Minkowski 1905 visited in Raper 2000 pg. 98)

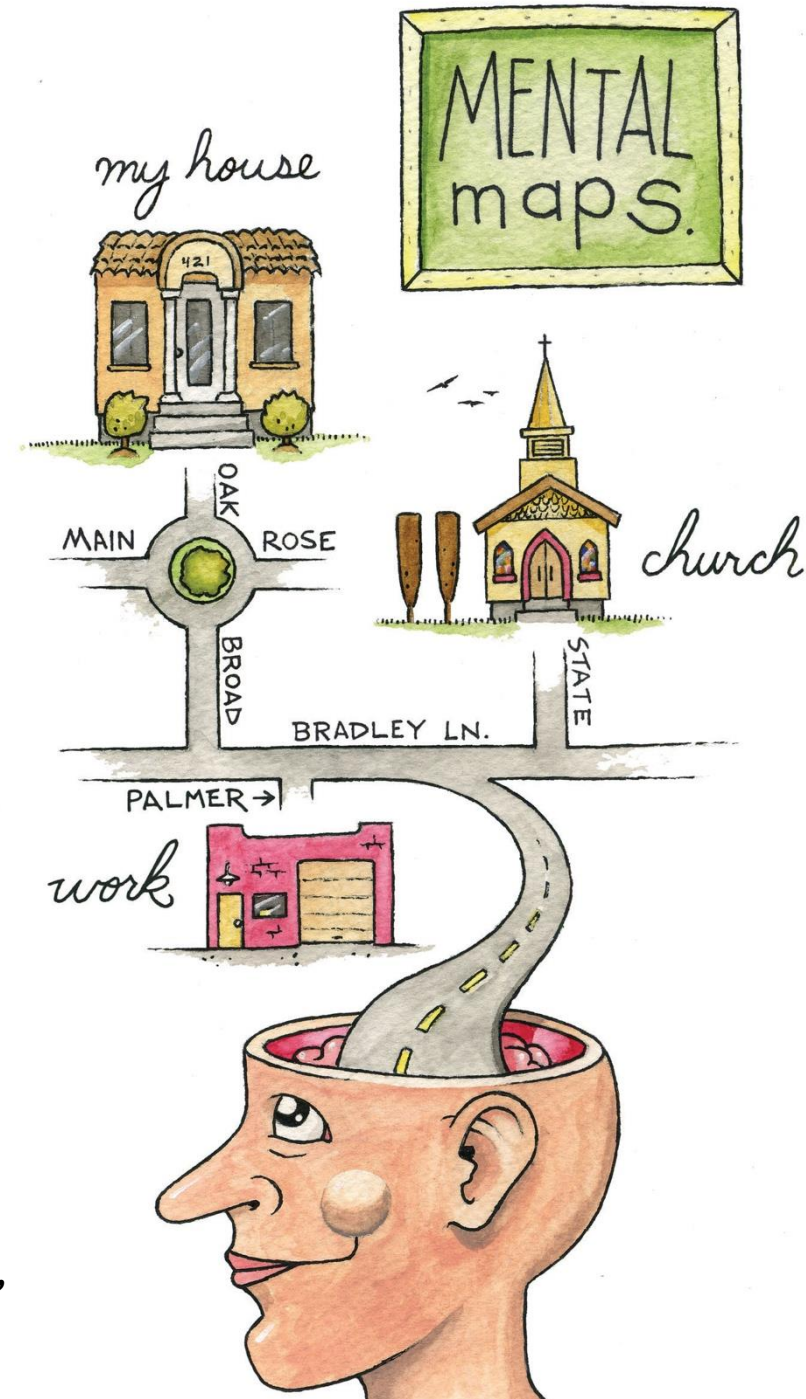
How do you view the movement of people over a day?



Mental maps

- We all have mental or cognitive maps of the environment we live in
- Now imagine that “Your friend is visiting IIIT, and you are directing him/her to reach to Vindya canteen”
- What if you want to draw the map for your friend?
 - What object will you consider?
 - How would you represent them?
 - Will you use any labels?
 - Will your map be similar to someone else’s map?

“Your mental map of a place represents your awareness about that place”



*“For each use case, decide whether the data is **non-spatial**, **spatial**, or **spatio-temporal**, and be ready to justify your choice”*

“A list of daily electricity bills for 5,000 households in a city for the year 2025.”

“GPS logs of city buses recorded every 10 seconds throughout the day.”

“A customer database containing their name, age, income, and total yearly spending.”

“Satellite scans a river basin every month during the monsoon season.”

“Police records of crime incidents capturing the locality of the crime, the exact date and time of occurrence.”

“A map of a city showing the land-use, created from a survey conducted once in 2018.”

Cholera in Soho District of London, in 1854



In 1854, in a single ten-day period, 500 people suddenly died in London, England.

All in the same neighbourhood.

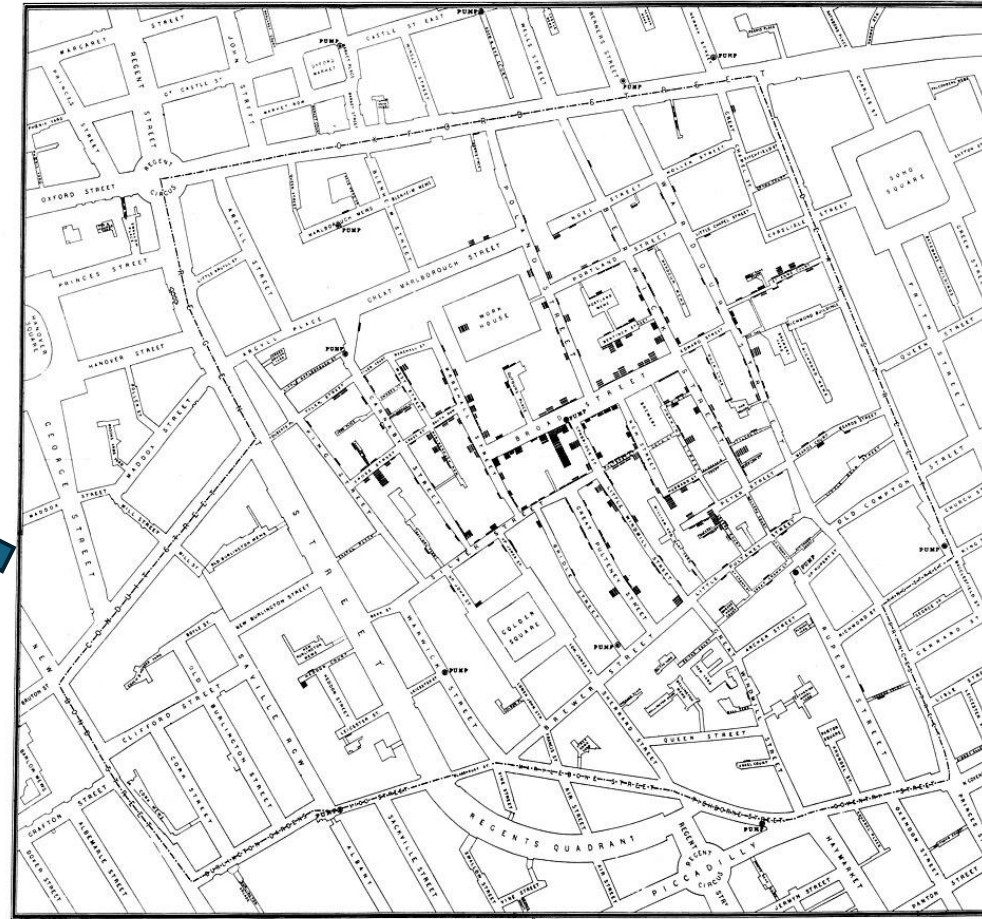
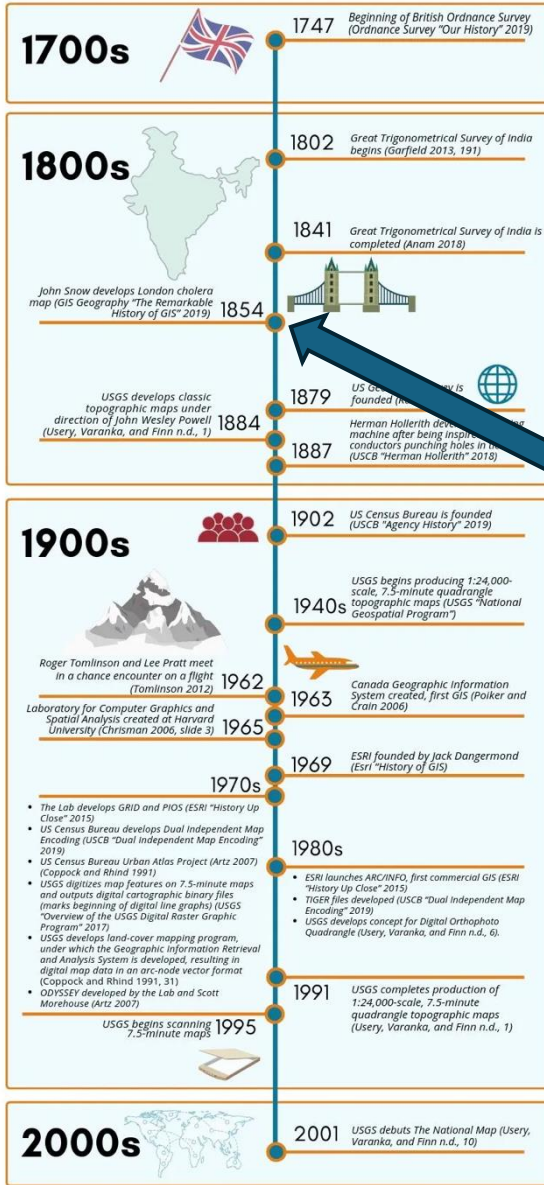
The cause of death was determined to be cholera, a severe infection of the small intestine.

What was the source of this disease?

How was it spreading?

A physician named Dr. **John Snow** was determined to find out.

KEY MOMENTS IN GIS



Dr. John Snow and the London cholera map

Use of spatial analysis in epidemiology, even way before the term GIS was coined.

A Geographic Information System for Regional Planning

R. F. Tomlinson

*Department of Forestry and Rural Development,
Government of Canada*

As a tool in its program of rural development, Canada is developing a computer-based information system for the storage and manipulation of map-based land data. The system and its capabilities are described.

Canada, like many countries, faces an immense problem in both understanding and guiding the development of its land, water, and human resources. One of the major agencies created specifically to implement policy to attack this problem is the Rural Development Branch of the Department of Forestry and Rural Development. A primary task facing this agency is to assemble social (demographic), economic, and land data for an integrated analysis to enable problems of rural development to be specified, development programs to be implemented, and their effectiveness evaluated.

Parallel with the gathering of data has been the development, by the Regional Planning Information Systems Division of the Branch, of interrelated computer-based information systems to handle and analyse the data. The Geographic Information System, for the storage and manipulation of land data is the most developed of these systems. Its design and development started in 1963, implementation began in 1965, and is now in its final stages; routine use is scheduled for September 1968. It is perhaps worthwhile to recount our progress with this system at this time.

Early in the life of the Branch (1962) a start was made with the gathering of some kinds of land data by the Canada Land Inventory. The data they collect is restricted to five types: the present use of the land, the capability of the land for agriculture, the capability of the land for forestry, the capability for recreation, and the capability for supporting wildlife. These data alone, if gathered in sufficient quantities for the summaries to be directly applicable to provincial and federal resource policy and regional planning, will generate an estimated 30,000 map sheets, at various scales. The Inventory has currently produced 7000 map sheets, of which 3000 have been prepared for computer input. The maps contain an average of 800 distinct areas on each sheet, and have been found to contain as many as 4000. Additionally, other types of maps covering watersheds, climate, geology, administrative boundaries, and land titles are generated by other agencies.

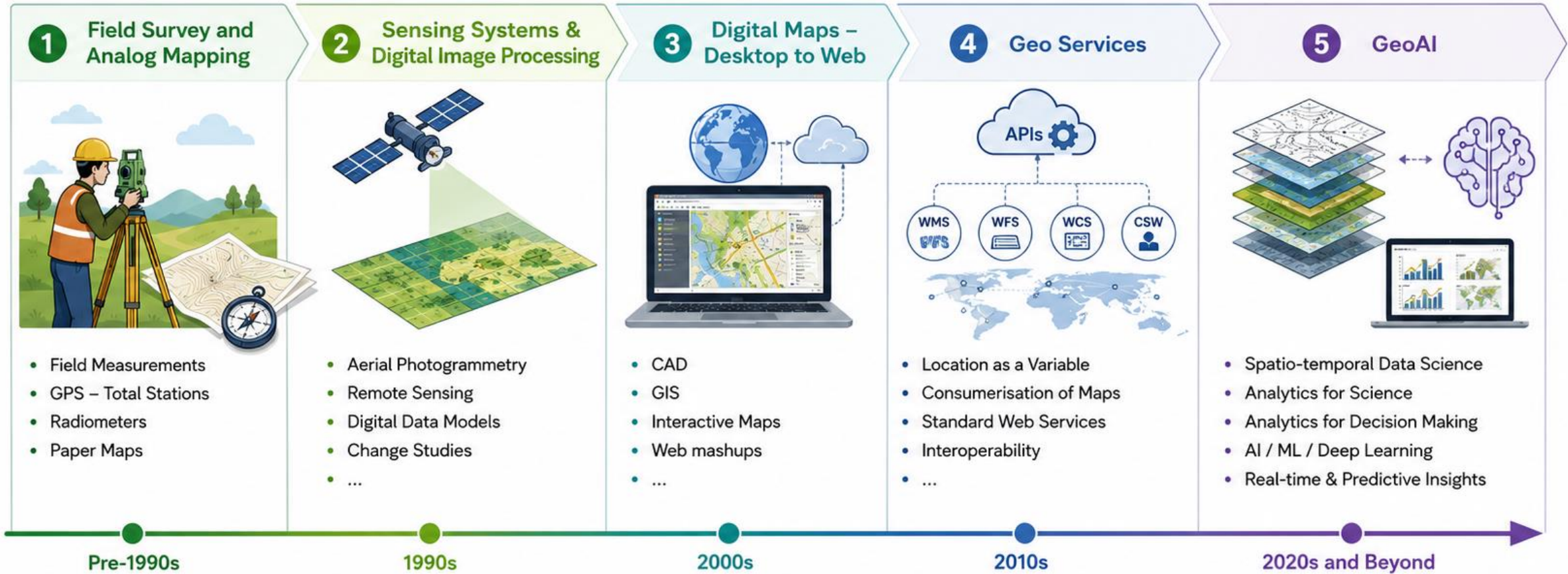
The need for a computer-based system, whereby map and related data can be stored in a form suitable for rapid measurement and comparison, is apparent as soon as the magnitude of the problem of handling large numbers of maps is appreciated. Lack of trained personnel makes it impossible to examine such large amounts of data manually in any sensible time, much less to provide a meaningful analysis of the content. A situation can be reached where the amount of data precludes its use. The end product of countless hours of survey can remain unused, with the result that administrators do not receive information necessary for a sound basis to decision making.



1960

- Roger Tomlinson known as the “father of GIS”
- Develops the First True Operational Geographic Information System (GIS)
- Tomlinson first published the term "geographic information system" in his paper entitled "[A Geographic Information System for Regional Planning](#)"

EVOLVING WORLD OF GEOSPATIAL TECHNOLOGY



Geospatial Landscape in India

From Mapping the Nation to Intelligent Geospatial Solutions

MAJOR PLAYERS



1. Cartography and Mapping

- Survey of India (since 18th century)
- Topographic maps, cadastral maps
- National Mapping Organisation (NMO)



Survey of India



2. Aerial Photogrammetry and Satellite Technology

- ISRO – Satellite remote sensing missions
- NRSC – Data acquisition and processing
- Bhuvan, Resourcesat, Cartosat, RISAT



3. Solution Providers

- Multiple agencies: GSI, FSI, NIC, etc.
- Private companies and startups
- Academia and Research Institutions
- R&D in geodesy, surveying, GIS, remote sensing, and more



India's Geospatial Journey



Enabling Governance,
Development and
Sustainable Future

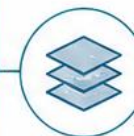


TECHNOLOGY EVOLUTION



1. Aerial and Satellite Image Processing

- From data acquisition to information extraction
- Orthophotos, DEMs, Land Use/Land Cover
- Map as a Product



2. GIS as a System

- From mapping to spatial analysis
- Integrating spatial and attribute data
- Decision support for planning and management



3. Information Technology

- Static to Interactive Maps (WebGIS)
- Cloud, Mobile and Real-time Mapping
- GeoSpatial Services and APIs



4. Spatial Data Science and GeoAI

- Geospatial data as a science
- Machine Learning, Deep Learning, AI
- Big Data, IoT and Real-time Analytics
- Smart Cities, Digital Twins, 4IR Applications



Key Enablers



Space Program

ISRO Missions



Digital India & Data Policy

Open Data, NGDI



Infrastructure

GNSS (NavIC),
HPC, Cloud



Capacity Building

Education, Training,
Research



Public-Private Collaboration

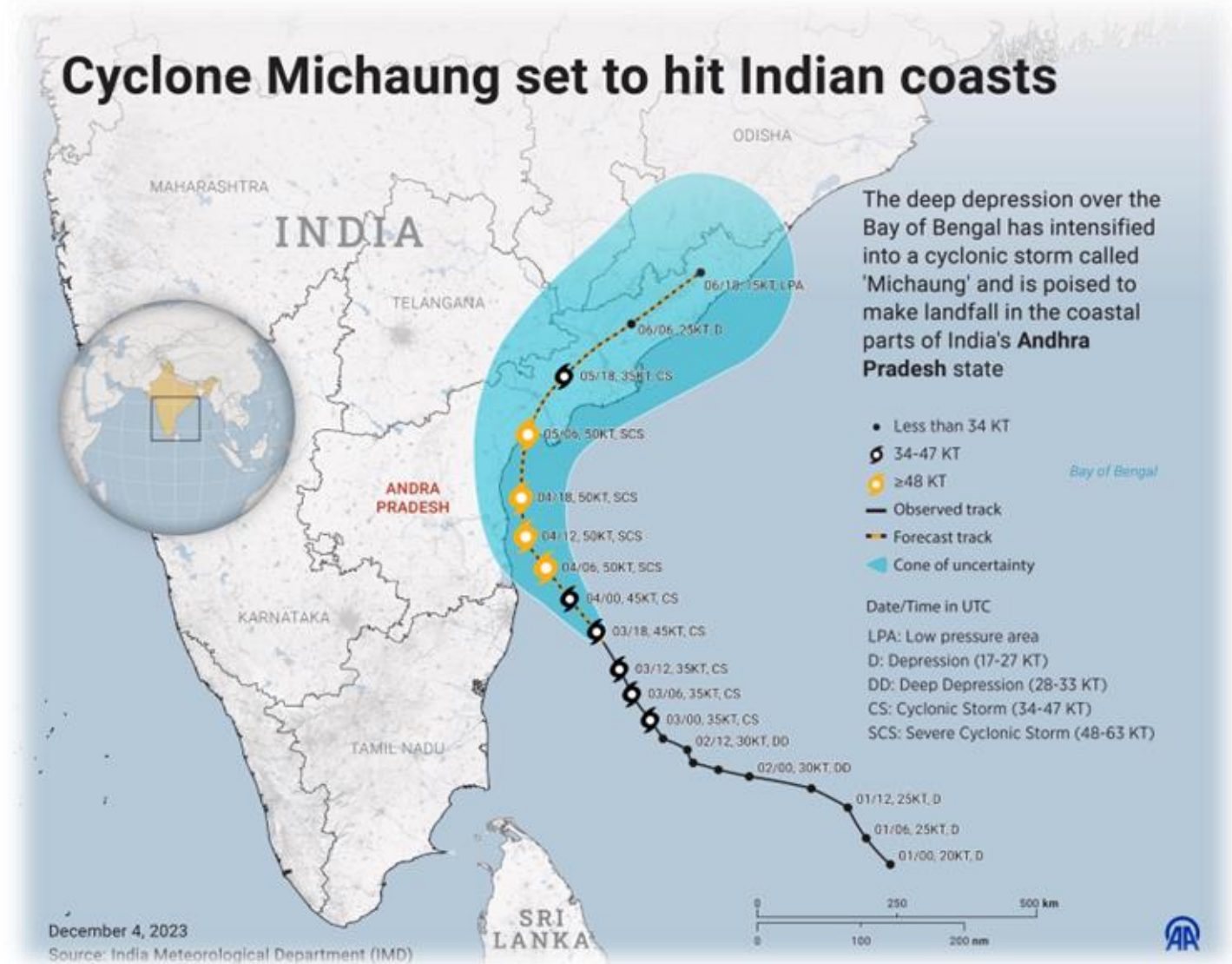
Innovation & Solutions



Geospatial Information – The Foundation for Informed Decisions and a Viksit Bharat

Location is important

- Something always happen somewhere
- Knowing where something happened can help understand
 - What happened?
 - When it happened?
 - How it happened?
 - Why happened?



Key Geographic Questions

Geographic location:

- Where is it?
- Why is it here or there?
- How much of it is here or there?

Geographic distribution:

- Is it distributed locally or globally?
- Is it spatially clustered or dispersed?
- Where are the boundaries?

Geographic association:

- What else is near it?
- What else occurs with it?
- What is absent in its presence?

Geographic interaction:

- Is it linked to something else?
- What is the nature of this association?
- How much interaction occurs between the locations?

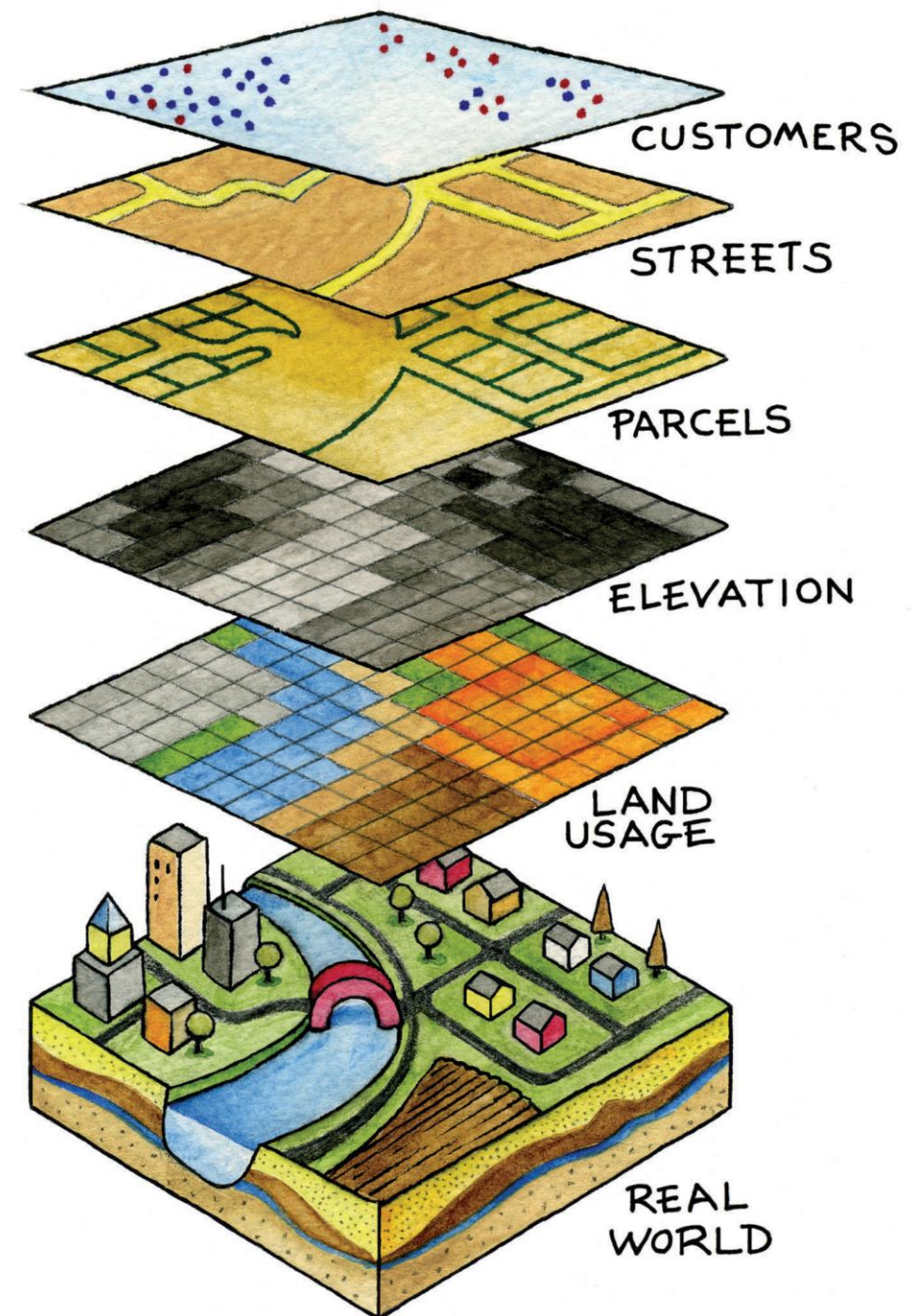
Geographic change:

- Has it always been here?
- How has it changed over time and space?
- What causes its diffusion or contraction?

“GIS helps to answer these types of questions”

Map Layers (Layered architecture)

- Various themes from real world can be represented as individual layers
- Represented as overlay of a set of layers



"Geographic Space"



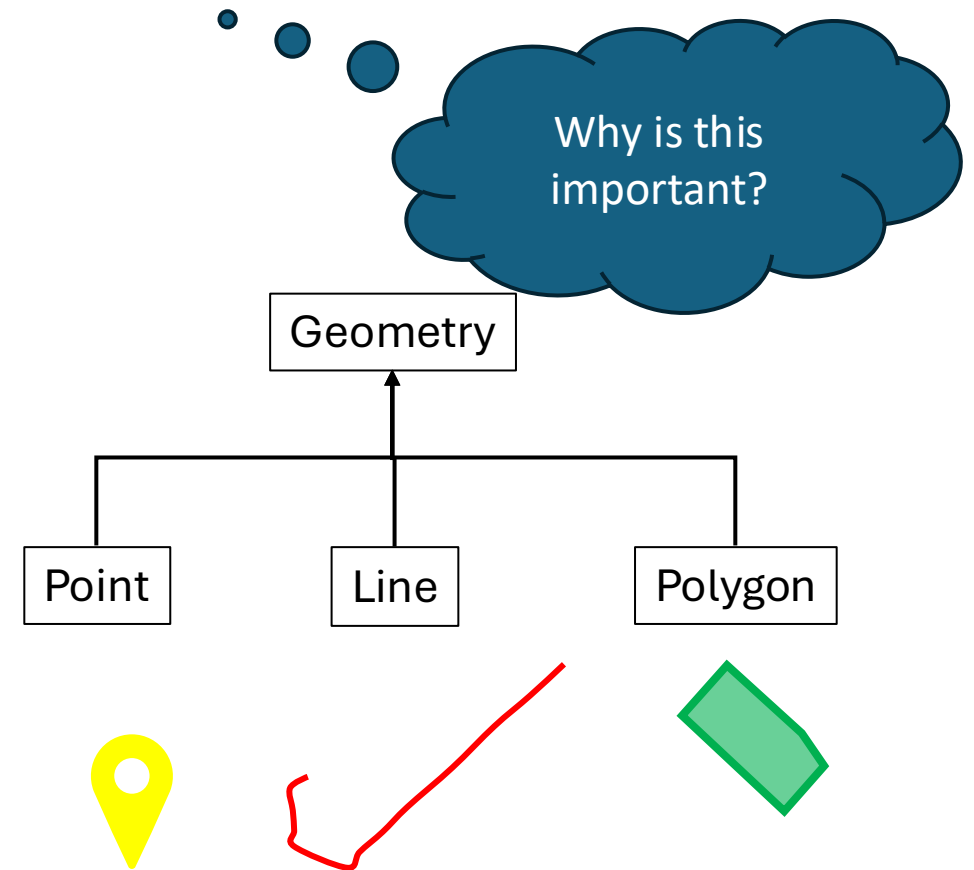
Use your best judgment:

- What does this Geographic space represent?
- Name the marked entities.
- How are these entities represented?
- How are these entities related (spatially) to each other?



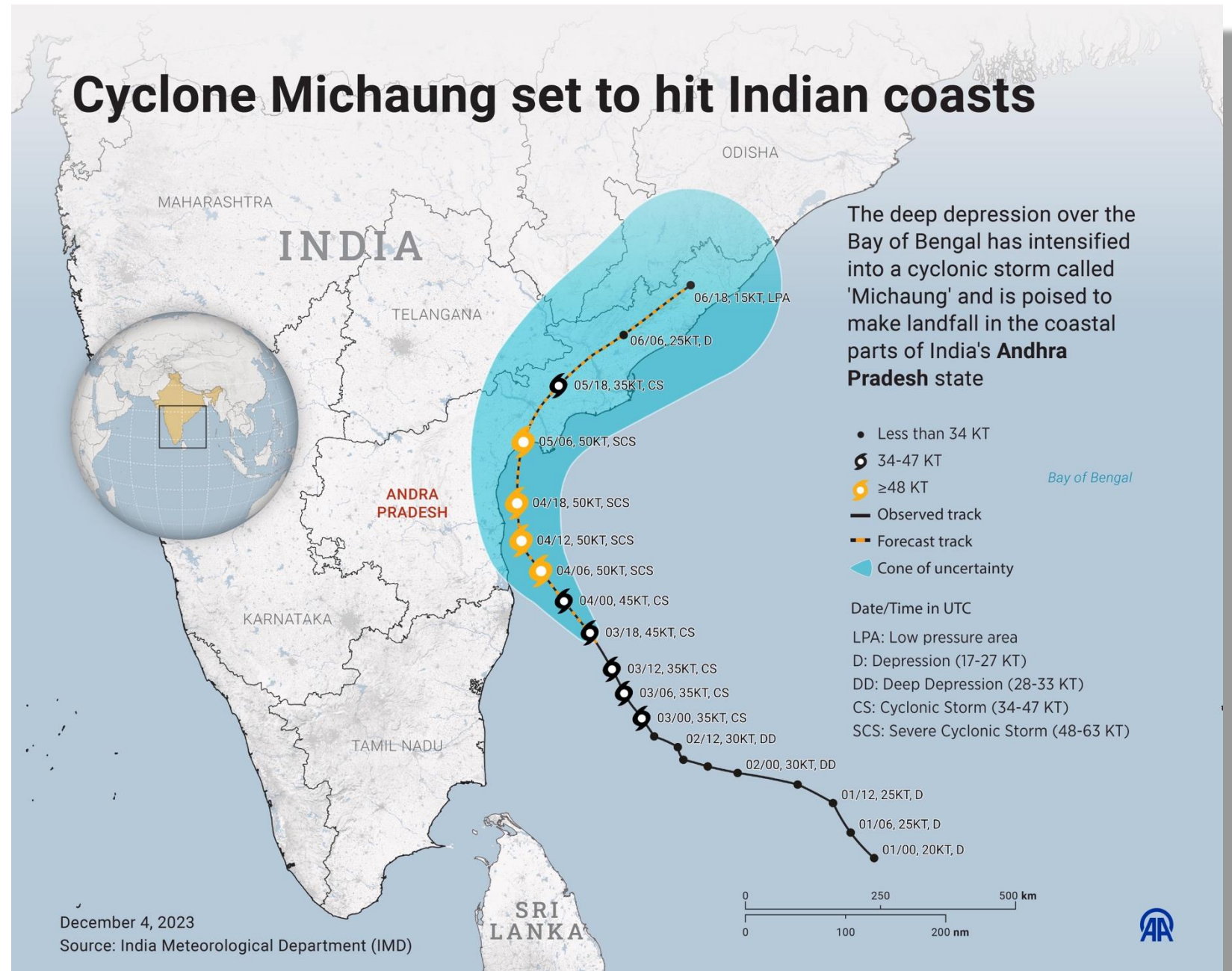
Use your best judgment and

- Do you agree with the representation?
- What could be the digital representation of these entities?



Fundamental concept underlying these representations is that of *geometry*

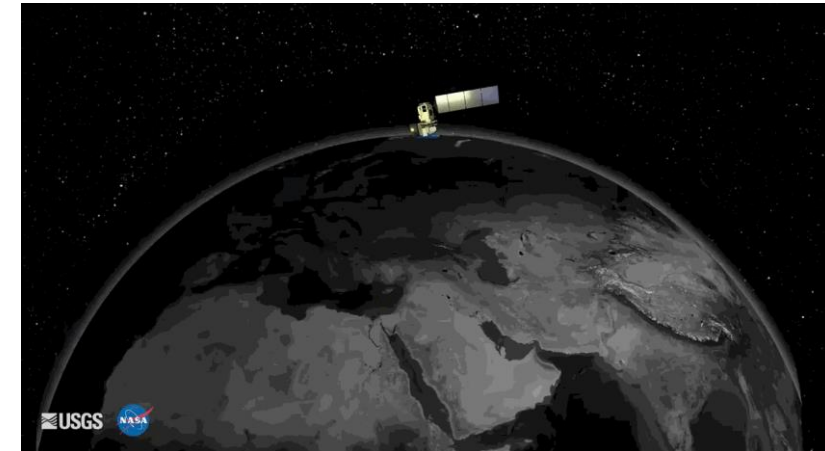
What are the various map components do you see in this map?



Various sources of spatial data

1. Remote Sensing (Satellite & Aerial)

- Satellite images from Sentinel or Landsat used to map floods, crops, or urban growth.
- Drone imagery for high-resolution mapping of construction sites or disaster areas.
- Key idea: Synoptic view, repeated observations, large-area coverage.



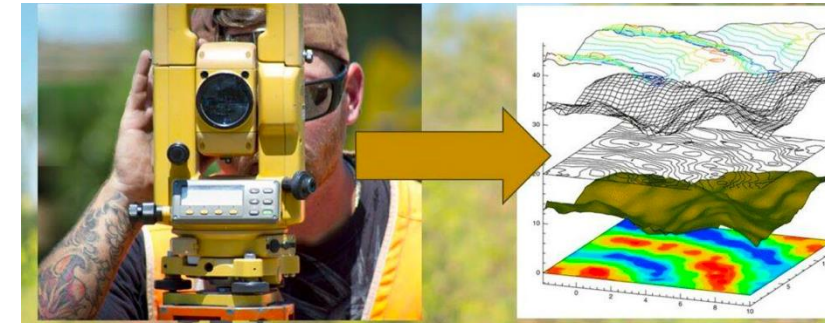
2. Global Navigation Satellite Systems (GNSS / GPS)

- GPS tracks from mobile phones, vehicles, delivery fleets, or wildlife collars.
- Fitness apps recording running or cycling paths.
- Key idea: Point locations, trajectories, high temporal resolution.



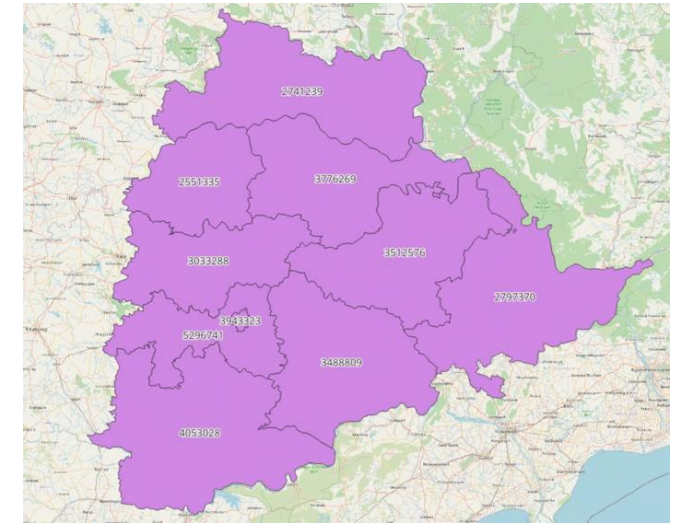
3. Survey & Field Data

- Locations of borewells, trees, or buildings collected using handheld GPS devices.
- Household surveys with geotagged responses.
- Key idea: High accuracy, but sparse and costly to collect.



4. Administrative & Government Data

- Census data aggregated at village, ward, or district level.
- Land records, cadastral maps, zoning maps.
- Key idea: Authoritative, structured, often aggregated spatially.



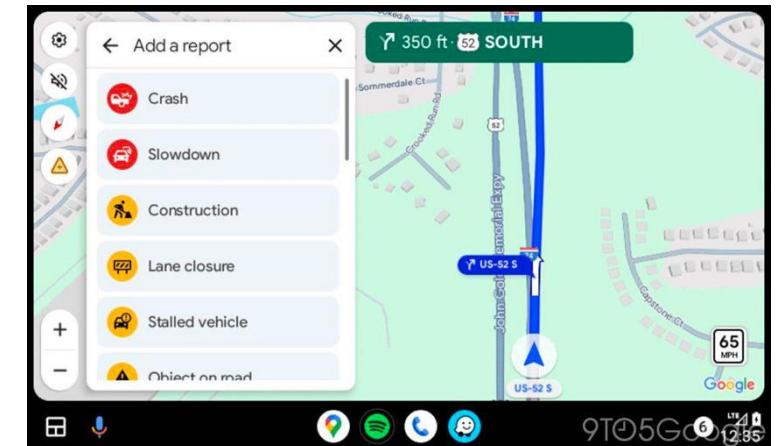
5. Sensors & IoT Networks

- Weather stations recording rainfall, temperature, and wind at fixed locations.
- Air-quality sensors deployed across a city.
- Key idea: Fixed locations + continuous time → spatio-temporal streams.



6. Crowdsourced & Volunteered Geographic Information (VGI)

- OpenStreetMap roads and buildings.
- Geotagged social media posts during events or disasters.
- Key idea: Large-scale, dynamic, variable quality.



What to do with the Spatial Data?



Primarily to **Visualize**
(**Depict or Draw**) the features

Location of a Spatial Features (Objects)

- Points like SBI ATM within IIITH
- Linear Objects like River Ganges
- District of Hyderabad



Locate or identify the interested
Spatial Object/Objects ?

Query the data to highlight

Extract the data (for use here or elsewhere)
Restaurants near me



What about **Analysis** ?

Spatial relationships ?

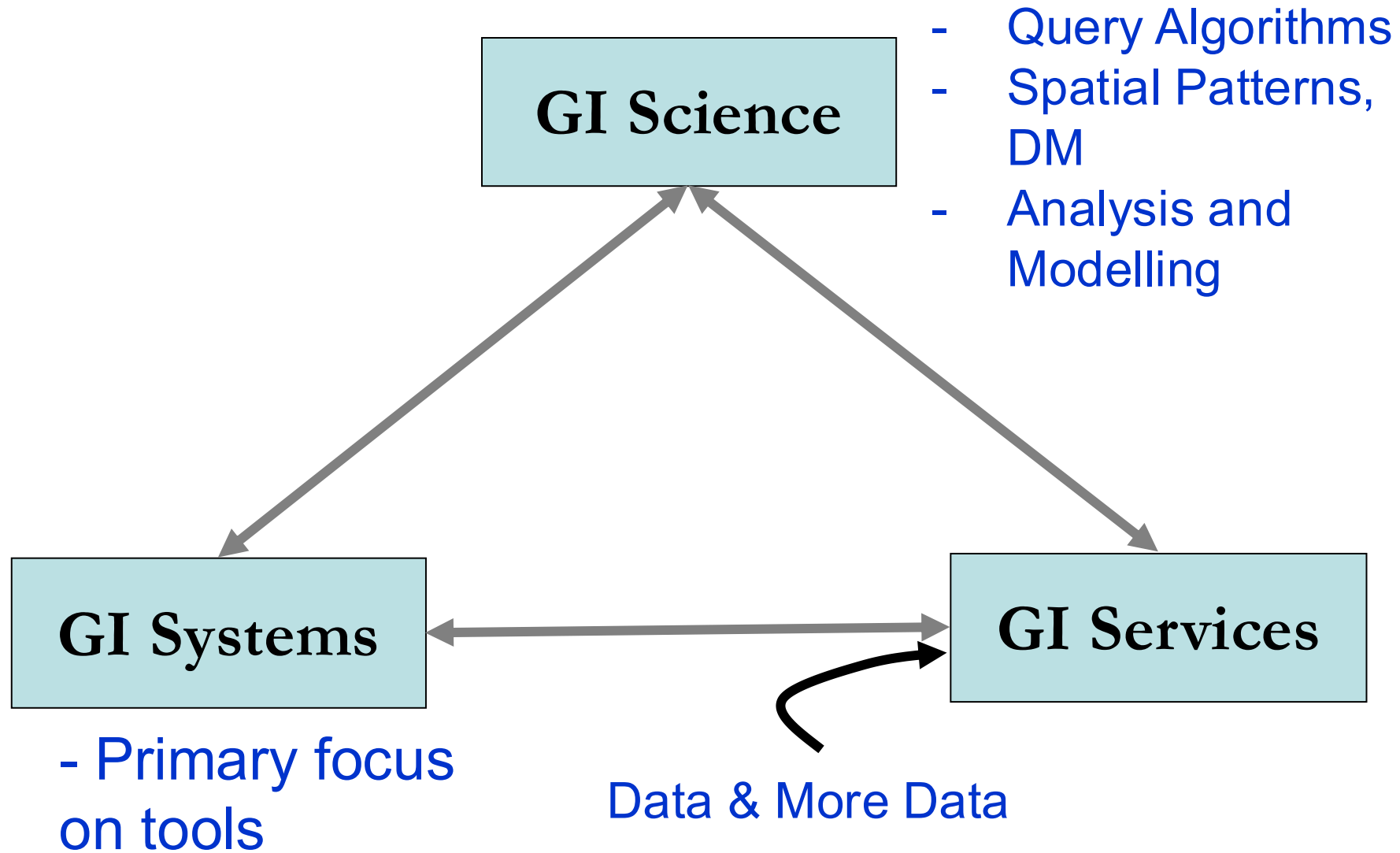
Cause-effect relations; Co-occurrence; mathematical
Urban Heat Island and its spatial factors



Spatial **Modelling** – what and
how?

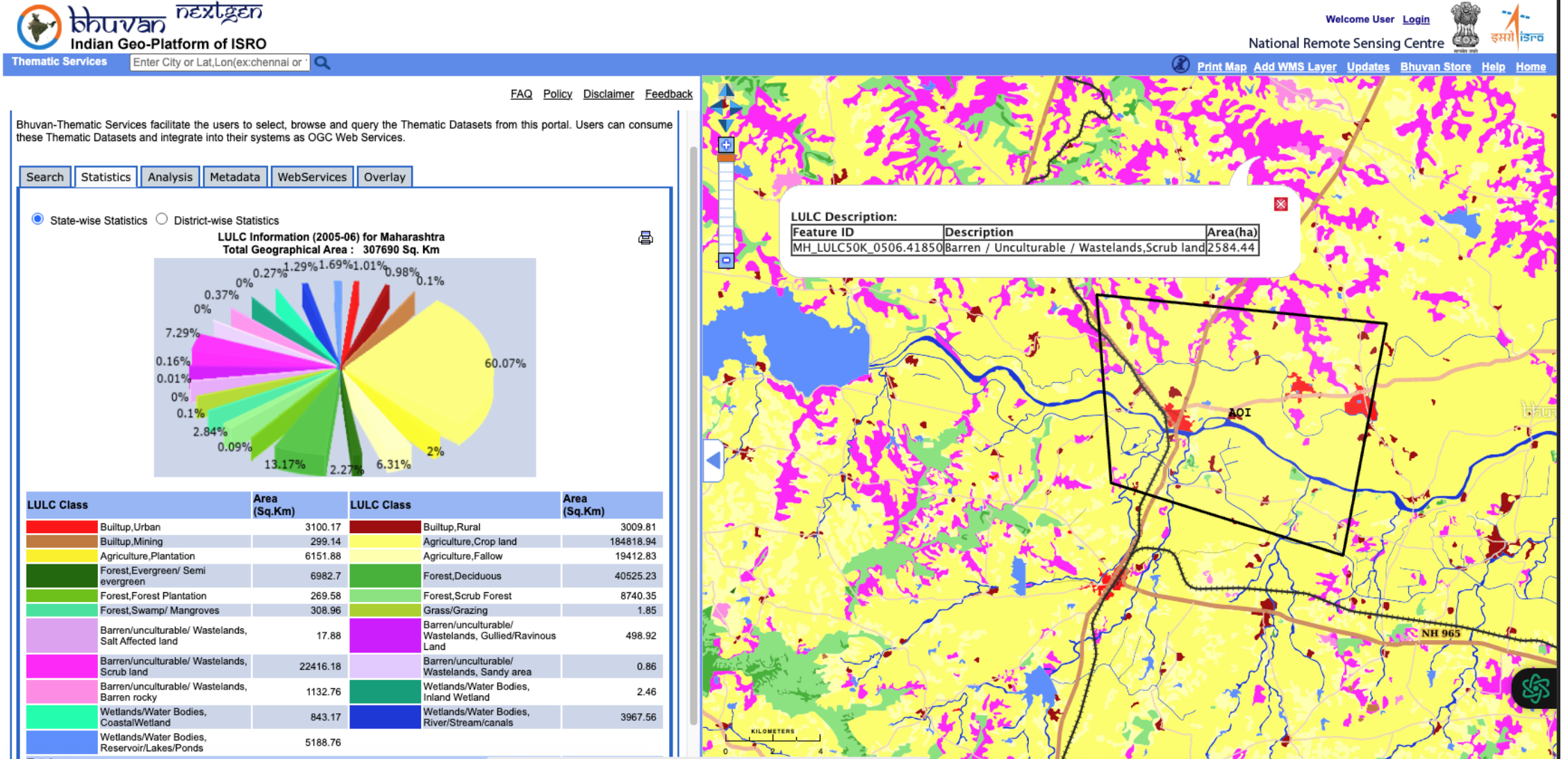
Disease spread modeling

Urban sprawl modeling



Modified based on Longley, Goodchild, Maguire, Rhind [2001]

Land Use Land Cover mapping



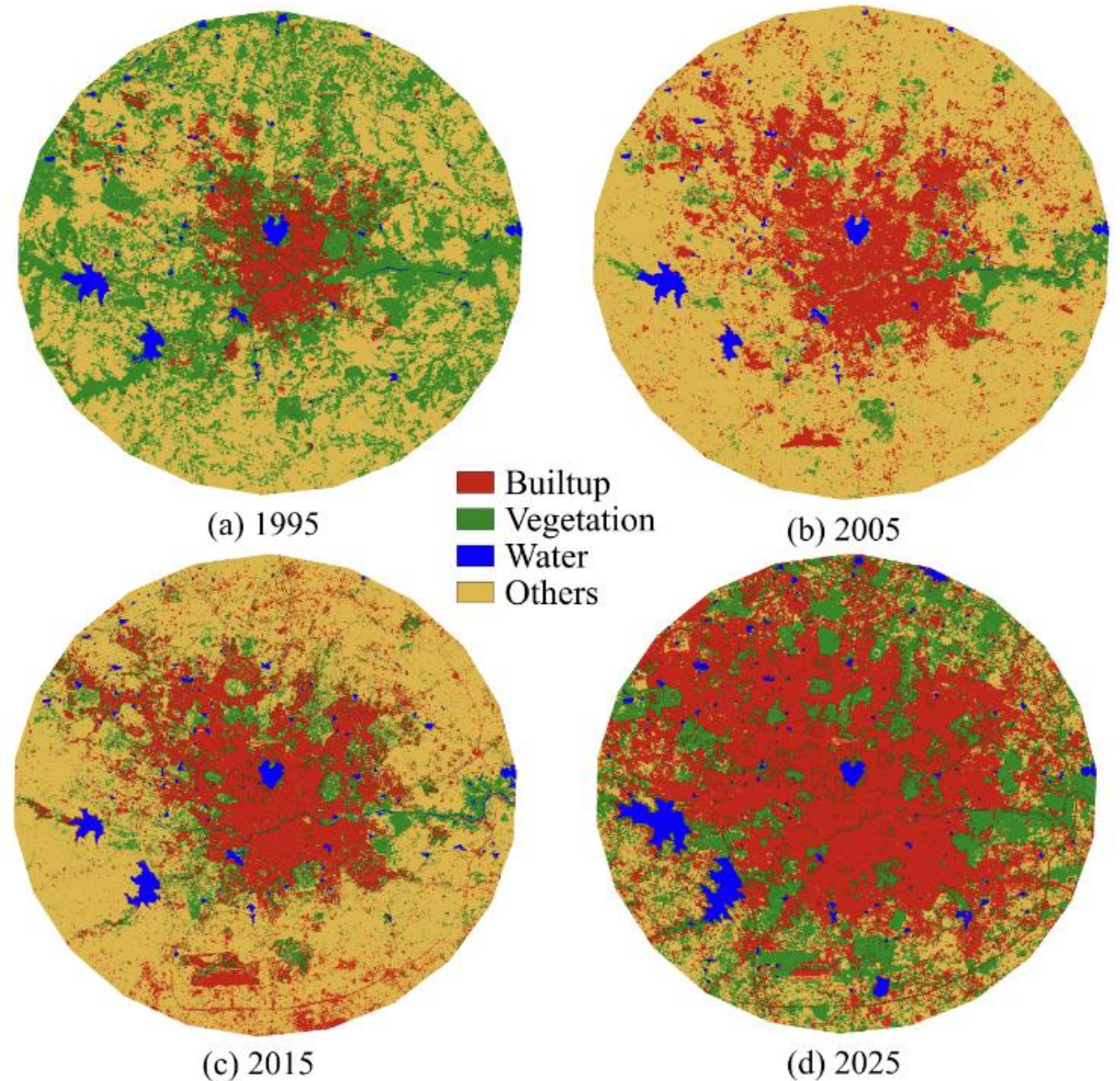
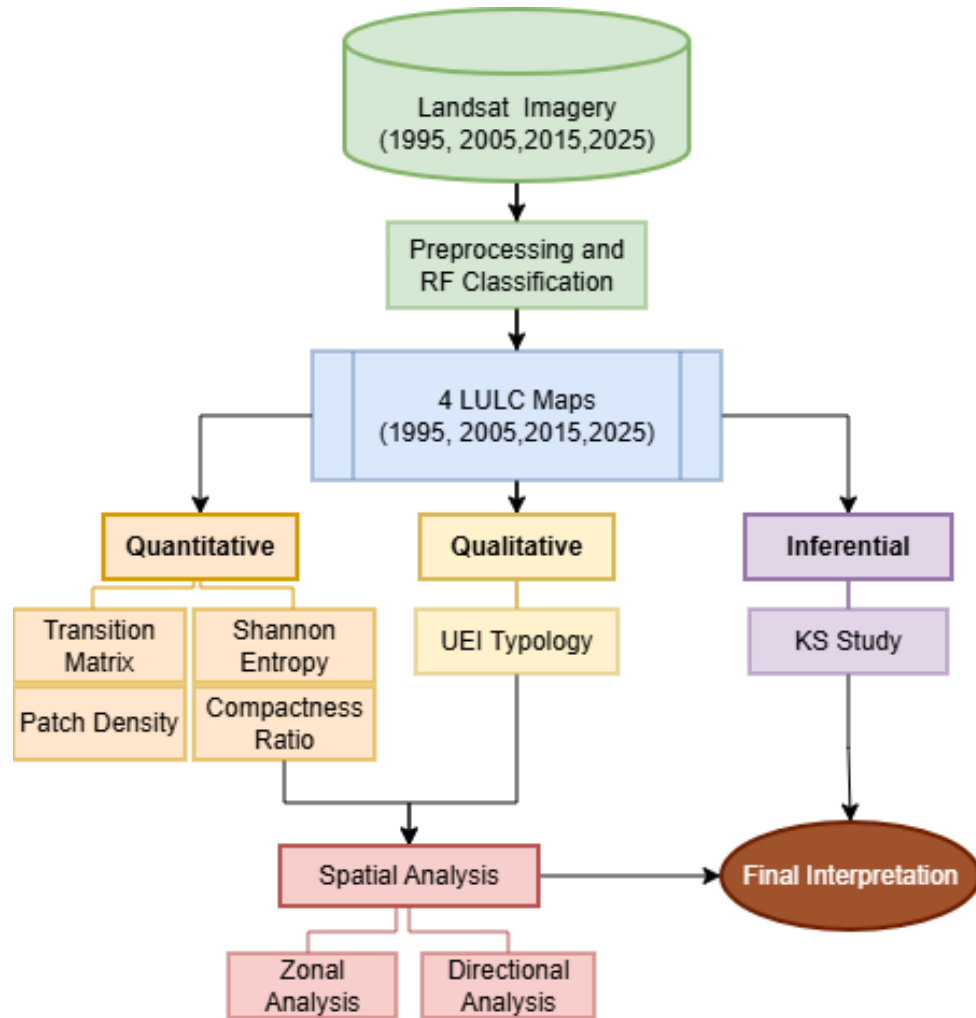
Bhuvan Web service: <https://bhuvan-app1.nrsc.gov.in/>



COVID-19 Dashboard by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University (JHU)



Urban Sprawl Analysis



How Collects Spatial Information

A real-world example of crowdsourced geospatial data collection through a game

1. Data Collection Sources



GPS & Location
Player's device GPS captures precise location, movement and visit patterns.



Camera & AR
The camera captures surroundings for AR gameplay and object detection.



Device Sensors
Accelerometer, gyroscope, compass and barometer help understand movement and environment.



Crowdsourced Player Actions
Millions of players contribute data through gameplay.



Every action you take in the game generates valuable spatial information!



2. How the Game Collects Spatial Data



Player Moves in Real World
GPS continuously records your location, movement, speed and direction.



Visits to Points of Interest (POIs)
When players visit PokéStops, Gyms or locations, the game logs the coordinates and visit frequency.



Scans Environment (AR)
Camera and AR features capture images of real-world objects, buildings and landmarks.



Tracks Interaction & Behavior
Data on where players catch Pokémon, battle, walk, explore and how long they stay at a place.



Data Sent to Niantic Servers
All collected spatial data (anonymized) is transmitted to servers for processing and analysis.

3. Types of Spatial Data Collected



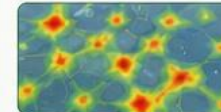
Location Points
Precise latitude, longitude, altitude and timestamp of player activity.



Visited POIs
Locations of PokéStops, Gyms, RAID locations and custom player destinations.



Scanned Real-World Objects
Images and spatial location of buildings, monuments, signs, murals, sculptures, and other features.



Movement & Density Heatmaps
Aggregated movement patterns and population density at different places and times.



Pathways & Mobility Patterns
Common walking routes, trails, commuting paths and exploration behavior.

4. How the Data is Used



Map Creation
Builds the game world map with accurate real-world locations.



POI Generation
Identifies and creates new PokéStops, Gyms and events.



Analytics & AI
Understands player behavior, popular areas and trends.



Fraud & Safety
Detects abnormal movement or cheating using spatial patterns.



Game Improvement
Improves gameplay, events, AR experience and location accuracy.



Player privacy is protected: Data is anonymized and used in aggregate form.

5. Impact: Beyond the Game



Crowdsourced Mapping
Helps create and update maps in areas where traditional mapping is difficult or expensive.



Disaster Response
Movement data can help in evacuation planning and impact assessment.



Urban Planning & Research
Provides insights on mobility, public space usage and urban dynamics.



Environmental Monitoring
Supports studies on green spaces, trails, tourism and habitat exploration.



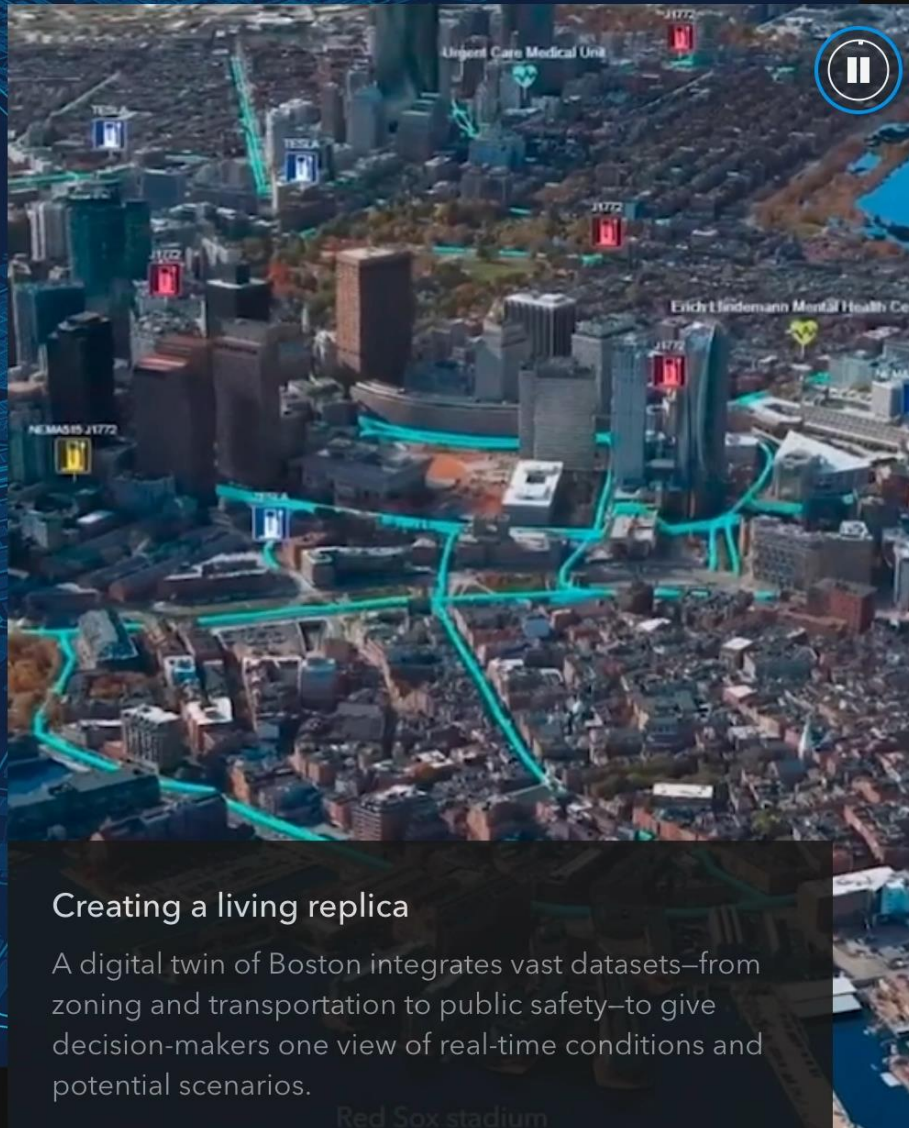
Education & Innovation
Encourages spatial thinking, data science and geospatial innovation.



Pokémon GO is more than a game – it is one of the largest real-time geospatial data collection platforms in the world, built through the power of millions of players exploring the real world.



NIANTIC



Creating a living replica

A digital twin of Boston integrates vast datasets—from zoning and transportation to public safety—to give decision-makers one view of real-time conditions and potential scenarios.

What is a digital twin?

Mirroring the real world with GIS

A digital twin is a virtual representation of reality, including physical objects, processes, and relationships. When built on a foundation of geography, it becomes a geospatial digital twin.

Geospatial

Accurate representations of complex systems require robust maps and spatial analytics from geographic information system (GIS) technology.

Time aware

GIS-based digital twins go beyond 3D models as they reflect change over time—showing historic, current, and future states.

Scalable

From single facilities to large built and natural systems, geospatial digital twins scale to meet changing needs.

Thank you